

Hiding a Message in a Voynich-like Book

The manuscript’s writing rules, measured against languages and medieval scribes,
and reproduced by a steganographic generator

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Preprint, version 2, October 2026. Not peer reviewed.

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Abstract

We present the *voynichizer*, a program that turns any text into a book of 207 folios written in the script of the Voynich manuscript (Beinecke MS 408), in EVA transliteration. The text is compressed, encrypted and authenticated with a key, and the resulting bits are sampled into the book through an arithmetic decoder: they decide only how many times each recurring Voynich word appears on each page, while layout, invented words and word order follow the manuscript’s statistics. With the key the text is recovered exactly; without it, a book that carries a message and a book that carries none are samples of the same distribution.

The generator is specified by a scorecard drawn from 716 logged experiments on how the manuscript is written (670 preregistered), each compared with the same measurement in 24 natural languages, constructed languages, volunteers’ gibberish, published generators and ciphers, and 79 medieval manuscripts (six Nordic, 73 German). The measurements quantify known regularities (word boundaries redundantly marked by word form, line-edge forms, paragraph-initial gallows, copying from the line above) and add others: the dependency between adjacent words mirrors the glyph sequences inside words and survives nulls that keep word form and line position; each line avoids starting like the line above; marked variants drift along the line at least two and a half times as much as in any scribal choice.

On 24 keys, two classifiers retrained on each generated book separate its pages from the real ones with AUC 0.55 and 0.60 (0.5 is chance; the real manuscript with its lines shuffled scores 0.60), and the book passes 16 of the 17 attainable scorecard properties. Statistical regularities of this kind, however strong, cannot by themselves decide whether the Voynich manuscript carries a message.

Keywords: Voynich manuscript; linguistic steganography; arithmetic coding; writing systems; medieval scribes; historical cryptography; preregistration.

1. Introduction

1.1 The manuscript and the question

The Voynich manuscript (Yale, Beinecke MS 408) is a codex of about 240 pages with drawings of unidentified plants, astronomical and zodiacal diagrams, female figures in pools, pharmacy jars and a long final section of text arranged in short paragraphs. Its vellum has been radiocarbon-dated to 1404–1438 at 95% probability (University of Arizona, 2011; Zyats et al., 2016). The script has never been read. Prescott Currier showed in the 1970s that the pages fall into two

varieties of writing, now called languages A and B, and that more than one hand wrote them (Currier, 1976); Lisa Fagin Davis later distinguished five scribes (Davis, 2020).

A century of proposed readings has not produced one that the scholarly community accepts. In recent years the debate has narrowed to a question that statistics seems well placed to answer: is the text a meaningful message (a natural language, perhaps enciphered) or a meaningless product of a procedure? Some studies find the text statistically close to meaningless writing (Gaskell and Bowern, 2023; Rugg, 2004; Schinner, 2007; Timm and Schinner, 2020); others show that historically plausible ciphers can produce Voynich-like text from Latin or Italian (Bowern and Gaskell, 2023; Greshko, 2025; Kinnison, 2026).

1.2 What statistics can decide

Every statistical study of the manuscript measures regularities. This paper asks whether regularities, however many and however strong, can decide the question at all. Steganography gives a precise answer in principle: if a text that carries a message and a text that carries none are samples of the same distribution, no test on the text can tell them apart (Cachin, 1998; Hopper et al., 2002). The question then becomes constructive. Can one build a book that reproduces the measured writing rules of the Voynich manuscript and that carries, or does not carry, a message, with the two cases indistinguishable by construction?

1.3 Contributions

1. **The voynichizer** (Sections 8 and 9). A program that turns any text into a 207-folio Voynich-like book in EVA transliteration and recovers it exactly with the key (Caniatti, 2026b). The text is compressed, encrypted and authenticated; its bits drive an arithmetic decoder that samples how many times each recurring Voynich word occurs on each page. Everything else (layout, invented words, word order) follows the manuscript’s statistics and carries no information. To our knowledge, no earlier tool combines a cover model of a whole historical book, distribution-preserving embedding and authenticated encryption, evaluated against the measured properties of that book.
2. **A measured specification** (Sections 5–7). The scorecard and the judges that the generator must satisfy come from 716 logged experiments on how the manuscript is written. Each measure is compared with the same measure in 24 natural languages, constructed languages, volunteers’ gibberish, published generators and ciphers and, systematically, 79 medieval manuscripts transcribed at the level of letter forms. Many of the regularities were observed before, and we cite them property by property (Section 2); our contribution there is to quantify them with the same code against the same comparison texts and to add a few properties that we have not found in the literature.
3. **A consequence** (Section 10). The generated book carries an encrypted message and comes close to the manuscript on the measures we know how to make. The regularities of the Voynich text therefore constrain how it was written, but they cannot by themselves show whether it says anything.

The full research record is public: preregistrations, code, results, the laboratory notebook and a log of all experiments with their outcomes (Caniatti, 2026a). Appendix C lists the provisional readings that later checks corrected, including those prompted by an external review of the first version of this paper.

2. Related work

2.1 Statistical regularities of the script

Glyphs and words. The low conditional entropy of the next glyph was noted by Bennett (1976) and quantified against hundreds of texts by Lindemann and Bown (2020). Words follow a rigid internal grammar (Stolfi, 2000; Zattera, 2023); Zattera (2023) also found “separable” words that split into two attested words. Feaster (2020) predicted mid-line spaces from glyph adjacencies with over 95% accuracy and observed that uncertain spaces fall where his rules are inconsistent; Rozanova and Temerev (2026) describe two regimes of spaces, show that uncertain spaces are physically narrower in the images and predict spaces from the flanking glyph pair with AUC 0.98. Matlach et al. (2022) revisit the positional roles of glyphs.

Adjacent words. Currier (1976) noted that in language B the end of a word influences the beginning of the next, with *qo-* words following words in *-y*. Emma May Smith described the preference for *o-* forms after words in *-n* and related transformations (Smith, 2017a,b), and Smith and Ponzi (2019) measured the dependency between last and first glyphs across spaces (*y.q* and *n.o* frequent across spaces and almost absent within words), concluding that word breaks are real boundaries. Reddy and Knight (2011) found that the last letters of a word help predict the first letters of the next; Caruana et al. (2023) and Parisel (2026a) studied dependencies between adjacent words. Feaster (2023) warned that part of the *y.q* and *n.q* anomalies might come from position in the line.

The line. Currier (1976) called the line a functional unit, noted line-final symbols and observed no repeated sequence across a line break. Vogt (2012) measured word-length effects along the line; Smith (2015, 2016a) described *s-* and other forms added at line start, and Smith (2016b) proposed *-m* as a line-final form of *-r*. Feaster (2021, 2023) defined rightward and downward indices of glyphs (for example *qo-* lies further left than *o-*, *Sh* further left than *ch*); Bown and Lindemann (2021) summarise line and paragraph effects, and Rozanova and Temerev (2026) find the link between glyphs at independence across line breaks.

Lines above and below. Timm (2014) and Timm and Schinner (2020) observed that many words resemble words written shortly before, often in the lines just above, and built a generator on this idea. On the left margin, participants of the Voynich Ninja forum reported that a line-initial glyph tends to avoid the glyph directly above it (tavie, 2024).

Paragraphs, pages, hands. Paragraph-initial galls, especially *p*, are well known (Currier, 1976; D’Imperio, 1978; Feaster, 2023; Stafford, 2023). Davis (2020) identified five hands, and Davis (2025) noted that in the herbal section hands follow the bifolia, which she calls highly atypical; the current binding is not the original order. Currier’s languages have been confirmed with modern methods (Farrugia et al., 2023; Parisel, 2026b). Immediate repetition of words has been measured by Boxer (2023) (about 1% of word pairs) and Bown and Lindemann (2021).

2.2 Generators and ciphers

Meaningless Voynich-like text has been produced with a Cardan grille (Rugg, 2004; Zandbergen, 2021), with the self-citation procedure of Timm and Schinner (2020), with Trithemius’ word tables (Hermes, 2023) and with recent generators that target statistical fingerprints: two

amateur generators (Whitehatnetizen, 2026) and *voynich-fingerprint* (Sachak, 2026), which reproduces 44 statistics on held-out folios. On the cipher side, Bower and Gaskell (2023) show that word-level metrics are compatible with some kinds of encipherment, and the Naibbe cipher (Greshko, 2025) turns Latin or Italian into Voynich-like text by hand; its author presents it as a proof of concept, not as the method used for the manuscript. Kinnison (2026) shows that Naibbe ciphertext reproduces the manuscript’s positional entropy profile.

2.3 Steganography and its evaluation

Transforming a text so that it imitates the statistics of another goes back to the mimic functions of Wayner (1992). Cachin (1998) defined the security of steganography as the distance between the distributions of covers with and without a message, and Hopper et al. (2002) gave provably secure constructions. Feeding encrypted bits to an arithmetic decoder in order to sample from a language model, so that the message-carrying text follows the model’s distribution, is the method of Ziegler et al. (2019); Kaptchuk et al. (2021) made it cryptographically secure. Training a classifier to separate two samples and reading its accuracy is a classifier two-sample test (Lopez-Paz and Oquab, 2017). The voynichizer applies these ideas with a cover model of one specific historical book and is evaluated against measured properties of that book.

3. Data

3.1 The Voynich manuscript

We use three transliterations in IVTFF format (Zandbergen, 2025a,b): Zandbergen–Landini (ZL3b, EVA alphabet; our reference), Takahashi (IT2a, basic EVA) and Glen Claston (GC2a, v101 alphabet, which segments some glyphs differently). They are three largely independent readings, but they share the locus scheme curated by Zandbergen: line segmentation and page metadata are the same in all three. Replicating a line-level result on the three files therefore does not replicate where a line ends.

Treatment of the transliteration. We use the running paragraph text: 207 pages, about 4,100 lines and 34,900 words; labels, circular and radial text enter only the tests that say so. Uncertain readings ([a:b]) take the first option; ligature braces are removed; rare glyphs (@nnn;) and illegible glyphs (?) make a word unusable, and such words (0.6%) are dropped, so that their neighbours become adjacent (about 1% of adjacent pairs are affected). The uncertain space (EVA comma) counts as a space unless stated; the tests on uncertain spaces treat it separately. The EVA sequences *ch*, *sh*, *cth*, *ckh*, *cph* and *cfh* count as single glyphs. Pages, bifolia and quires are taken from the IVTFF page headers: 50 bifolia and 16 quires carry text. Hands follow Davis (2020) and languages follow Currier (1976), both as recorded in the files page by page.

3.2 Comparison texts

- **Natural and constructed languages.** The publicly released texts of Gaskell and Bower (2023): 71 texts, of which 56 are in 24 natural languages (usually a New Testament and a technical text per language, historical and modern; Latin also in an abbreviated form) and 15 in eight constructed languages (Esperanto, Interlingua, Klingon, Lojban, LOLCat, Neo-Quenya, Toki Pona, Volapük; LOLCat is a register of English rather than a constructed language). Eleven of the technical texts are translations of the Wikipedia article on the

manuscript; they contain at most one or two EVA words each. The collection also provides the texts we cite by name (the Qur'an, Avicenna, the *Hatton Gospels*, Alberti's *Della Pittura*, Pliny). In these files the lines are the original text cut every six to eight words, not physical lines. Comparisons use the first 10,000 words of each text and, since version 2, one value per natural language (the median of its texts), with constructed languages reported apart (experiment e3c84).

- **Other texts:** the Bible in about 100 languages (Christodouloupoulos and Steedman, 2015); Latin texts of the Latin Library (CLTK collection), used for the five-gram model of the cipher solvers; the Roman breviary (Divinum Officium), used in the tests on prayers and litanies.
- **Handwritten gibberish:** the 38 released texts written by volunteers for Gaskell and Bowern (2023).
- **Published generators and ciphers:** the Naibbe cipher (Greshko, 2025); the self-citation generator of Timm and Schinner (2020); the generators U2 and U3 of Whitehatnetizen (2026).
- **Medieval manuscripts transcribed at the level of letter forms**, with original lines and pages: six Nordic manuscripts from Menota (Menota) (AM 519 a 4to, Icelandic, c. 1280; AM 677 4to, Icelandic, c. 1200–1225; AM 60 4to, Norwegian, c. 1320; AM 242 fol, Icelandic, c. 1350; Holm A 10, Swedish, c. 1500–1550; AM 302 fol, Norwegian, c. 1300) and 73 German manuscripts of 1350–1500 from the Reference Corpus of Early New High German (Wegera et al., 2021). These are 79 manuscripts, not 79 known individual scribes: the files do not mark changes of hand.
- **A historical cipher:** the Copiale, a mid-eighteenth-century German homophonic cipher of 105 pages, with the transcription and decipherment of Knight et al. (2011a,b).

Sources, versions and licences are listed in Appendix E. All corpora except the Voynich transliterations are kept outside the repository and must be downloaded from their sources.

4. Methods

4.1 How the work was organised

Every experiment followed the same cycle, each step in a separate commit of a version-controlled repository:

1. the program was tried only on synthetic texts and controls, never on the Voynich;
2. a preregistration was committed: question, data, measure and the criterion that would decide the outcome;
3. the program was committed;
4. the experiment was run by a script that records the code version, data fingerprints, random seeds and output;
5. results and a dated entry of the laboratory notebook, which is only ever appended to, were committed.

The repository is public (Caniatti, 2026a). The preregistrations are internal: they are dated files committed before each run, but they carry no third-party timestamp. Appendix B summarises the log of all 716 experiments (e01–e3c92) with the outcome of each against its preregistered criterion; 698 were run, 670 were preregistered, 80 declare a deviation, and 19 of these are cases

in which the author read an outcome differently from the literal criterion (each listed with both readings).

Exploration and confirmation. The research programme was exploratory: each question arose from earlier results on the same text. Confirmation on data not used to formulate a hypothesis is limited to a few cases: the prediction that *qo-* would be almost absent from drawing labels, made before looking at them (it was already known from Zandbergen 2026); the replications on the Takahashi and Glen Claston transliterations; and the comparisons with the 73 German manuscripts, collected after the measures had been fixed on the Nordic ones.

4.2 Controls

- **Synthetic validation.** Every measure was first run on texts built with the effect present and absent. A measure that did not separate the two cases was not used; three experiments were dropped for this reason and declared as gaps (Section 11).
- **Positive and negative controls** in every decipherment test: a known text enciphered in the same way, which the method must solve, and a text in another language.
- **Nulls that keep what the hypothesis does not concern:** for example, word order shuffled within the line, choices shuffled among occurrences of the same word, or (since version 2) nulls that keep word form or position in the line (Section 5).

4.3 Main measures

The measures are described in words where they are used; their definitions are in Appendix A.

- **Junction:** mutual information between the last glyph of a word and the first glyph of the next word in the same line, minus its mean under a null that shuffles the words of each line.
- **Junction rule:** for a word that occurs in two forms (*qo-/o-*, *-l/-r*), the dependence of the form on the neighbouring word.
- **Agreement of choices:** for binary choices between variant forms, a kappa-like agreement between nearby words, corrected for the bias of the expected value by subtracting the agreement obtained when the choices are shuffled among occurrences of the same word.
- **Copying from the line above:** excess of identical or nearly identical words in the lines above over what the paragraph’s vocabulary predicts.
- **Left margin:** ratio of observed to expected identical line starts in consecutive lines, the expectation coming from shuffling the order of the lines within the paragraph.

4.4 Uncertainty and multiplicity

Intervals are bootstrap percentiles (2,000 resamples). In version 1 we resampled pages; but pages of the same bifolium are not independent (Section 5.7), so the main intervals are now computed by resampling bifolia (experiment e3c86). They are 0.84 to 1.56 times as wide as the page intervals, and every main measure keeps its interval clear of the null value; with only 16 quires, resampling quires widens them further and one drift (*k/t*) touches zero. For the main tests we declared seven families (one per finding) and applied Holm’s correction within each, and Bonferroni’s correction over the 715 experiments logged before this test as a stricter reference (e3c92): all 23 tests survive Holm, 15 survive Bonferroni (Table 7 in Appendix A). Effect sizes are given with their intervals; *z* values are given only as a guide.

4.5 Use of generative AI

The work was carried out in a continuous conversation with Claude, a large language model by Anthropic (mainly Claude Opus 5.5), used through Claude Code in September and October 2026. Claude proposed experiments, wrote the preregistrations, the analysis programs (Python 3.12) and the notebook entries, ran the experiments, summarised their outputs and drafted the text of this paper. The voynichizer was developed with the same system in a separate conversation on the same repository. The author set the questions and the direction of the research, chose among the avenues proposed, approved every data download and every change of scope, and reviewed results and text. The research conversation processed about 2.8 billion tokens, almost all of them the context of the conversation read again at each step.

5. The writing rules of the Voynich manuscript

This section describes how the text is written, from the word to the book. For each property we give the measurement, its interval and the comparison texts, and we say whether the property was known. Section 6 gathers the comparisons with languages, manuscripts and generators, and Section 7 turns the properties into the scorecard of the generator.

5.1 Word form and word boundaries

The next glyph is very predictable: the conditional entropy h_2 of the running text (space included) is 2.23 bits, against 2.54–4.65 in the 24 natural languages (as found by Bennett, 1976; Lindemann and Bower, 2020). Words follow a rigid internal grammar (Stolfi, 2000; Zattera, 2023), and four properties follow from it:

- spaces are predictable from the two glyphs on either side (F1 0.86; higher than 23 of the 24 languages, the exception being Chinese in pinyin), and uncertain spaces fall where this rule is uncertain (as observed by Feaster, 2020; Rozanova and Temerev, 2026);
- cutting the glyph stream at other places allowed by the rule mostly yields words attested elsewhere (50% of the wrongly cut pieces, against a median of 6.5% in languages; compare the “separable words” of Zattera 2023);
- the vocabulary fills the most probable glyph sequences (43% of them are Voynich words; median 5% in languages);
- frequent words are the most probable sequences (rank correlation 0.58 between form probability and frequency; median 0.09 in languages).

In version 1 we read these properties as evidence that spaces are weak boundaries inside a continuous chain of glyphs. That reading does not hold. A null that keeps the word grammar and removes every link between words reproduces all four (e3c85): with words drawn independently from the same vocabulary, or generated one by one from a glyph model of order two trained on Voynich words, the F1 of the spaces is 0.88, the attested wrong cuts 43–45%, the filling 42–43% and the frequency–form correlation 0.55–0.62. These properties show that word boundaries are marked redundantly by the shape of the words, as Stolfi (2000), Feaster (2020) and Rozanova and Temerev (2026) had described, and as in any script with positional forms. They do not show that spaces are weak.

5.2 The junction between adjacent words

A strong link. The last glyph of a word says a lot about the first glyph of the next: the junction is 0.188 bits beyond the in-line shuffle (95% interval by bifolium 0.163–0.211). Per language the median is 0.097; two languages of 24 exceed the Voynich, Sanskrit and Tagalog, which write phonological links between words (e3c84). The volunteers’ gibberish has 0.014, and published generators between 0 and 0.08. The link was observed by Currier (1976), Smith (2017a) and Smith and Ponzi (2019).

It mirrors the rules inside words. The preferences between the last glyph of a word and the first glyph of the next resemble the preferences between consecutive glyphs inside a word: the rank correlation between the two tables of pointwise mutual information is 0.47 (median of samples of 10,000 words), against at most 0.37 in the 24 languages and 0.39 in the constructed ones. This part does not follow from word form: the nulls of e3c85 bring it down to 0.13 or less (0.22 when only the order of the words within each line is shuffled), and the junction itself to zero. At equal h_2 the Voynich lies 4.6 standard deviations above the regression line of the languages. Smith and Ponzi (2019) concluded that word breaks are real boundaries, and Section 5.1 agrees; what we add is that the dependency across the boundary follows the same pattern as the dependency inside the word, more than in any language of the sample.

Two junction rules. (1) Before a gallows, a word begins with *qo-* after a word ending in *-y*, *-o* or *-d* (59–64% of the time) and with *o-* after *-n*, *-r*, *-s* or *-m* (*qo-* only 23–27%); in part this was described by Currier (1976) and Smith (2017b). (2) A word ends in *-r* before a word beginning with *a-* (85%) and in *-l* before *k-*, *t-*, *d-*, *l-*, *s-*, *q-* (63–86%). Both rules hold within every class of position in the line (e3c87): with a null that keeps position, the junction stays at 0.196 against 0.188, and in each interior position *qo-* follows *-y/-o/-d* 61–70% of the time against 23–33% after *-n/-r/-s/-m*. The caution of Feaster (2023) does not apply to these rules. They hold equally in the three main hands, in all sections and bifolia, and in the three transliterations. They have the shape of rules such as English *a/an* or French *liaison*; they say nothing about the sounds behind them, and they are compatible with a phonetically written language.

Looking one word ahead, and no further. When two words copied from the line above would jointly violate a rule, the piece that changes is the end of the first word for *-l/-r* (59% against 21%) and the start of the second for *qo-/o-* (60% against 18%). The pattern is compatible with writing that knows the next word while finishing the current one; it rests on few conflicts (37 and 60) and does not survive the strictest multiplicity correction. Beyond the next word there is no link: for a given middle word, the last glyph of a word says nothing about the first glyph of the word after next (excess 0.003 bits, $z = 1.3$), while 22 of 24 languages have such a link.

5.3 The line

The junction stops at the line break. Between the last word of a line and the first word of the next the link is zero (−0.003 bits, $z = -0.6$), against 0.19 inside the line; the ratio Q of the two is −0.02 on the whole text and 0.02 on samples of the length of the comparison texts (e3c89). In the comparison languages, whose lines are prose cut every six to eight words, the median Q is 0.49 and only Hebrew falls below 0.10. A herbal in verse, the *Macer floridus* printed one verse per line, has a closed line too ($Q = -0.004$). The line is therefore a unit of

the writing, as Currier (1976) and Rozanova and Temerev (2026) found; a line that is a unit of content (a verse, an entry, a recipe) would produce the same closure (Section 10).

And at the gap left by a drawing. Where a plant interrupts a line, the junction between the words on the two sides drops from 0.167 to 0.026 bits, and the word after the gap begins like a word at the start of a line.

Line-edge forms. For the same rest of the word, at line end *-m* gains 15 percentage points and *-r* loses 12 (*dar* inside a line, *dam* at its end); at line start *s-* and *y-* gain 9–10 points in place of *ch-* and *k-* (*z* from 13.6 to 26.7). In the last line of a paragraph, which does not reach the margin, *-m* is rarer: it is a form tied to the margin. These forms were described by Currier (1976), Smith (2015, 2016b) and Feaster (2020, 2023); we measure them for the same word and compare them with scribes in Section 6.2.

Labels. In the labels of the drawings, which stand alone, *qo-* before a gallows is almost absent (0–5% against 30–68% in lines). We predicted this from the junction rule before looking at the labels; the fact itself had been noticed by Zandbergen (2026).

5.4 The lines above

Copying from the line immediately above. A word has an identical or nearly identical word in the two lines above more often than the vocabulary of the paragraph predicts ($z = 17.4$), more than twice as much as in the generator of Timm and Schinner (2020), which was built on this observation of Timm (2014). What we add concerns the shape of the copying: it comes almost entirely from the line immediately above (in languages recurrence spreads over three or four lines); two neighbouring copied words come from two neighbouring words above, usually in the same order; a copied word takes the *qo-/o-* or *-l/-r* form that agrees with its new neighbour more than the form of its source (effect +0.49 against +0.16); and copying stops at the paragraph and at the page.

Each line avoids starting like the line above. Shuffling the order of the lines within paragraphs shows that a line avoids beginning like the line above: identical starts are 0.51 of the expected number (95% interval by bifolium 0.40–0.61; 0.53 on samples of 10,000 words), with the strongest effects for *qo-* ($z = -9.7$) and *o-* (-7.0). If the line above starts with *qo-*, the probability that the next starts with *qo-* drops by 52 points. The effect concerns the left margin only: it is absent where lines resume to the right of a drawing, and the line two above, the end of the line above and the right margin do not matter. In the languages the median ratio is 0.97 and one language of 23 falls below the Voynich (Portuguese, in two texts with artificial lines); the volunteers’ gibberish has 1.32 and the generators 0.93–2.05. A forum study had reported vertical avoidance of the same glyph in the herbal pages of one scribe (tavie, 2024); we measure it across the manuscript and against the comparison texts.

5.5 Choices between variant forms

Many Voynich words exist in pairs that differ by one glyph: *qo-/o-* at the start, *k/t* (two gallows), *sh/ch*, *-ey/-dy* at the end. We treat these as *variant forms*: choices whose distribution we can measure for the same rest of the word. This is a working hypothesis, not a result. If

the variants were morphemes, their agreement between nearby words would be a kind of grammatical agreement, which languages have; we therefore compare them both with grammatical agreement in languages and with allographs in medieval scribes.

A short state. Between nearby words of the same line, the same choice is repeated more than the preferences of each word, page and position predict. With the bias of the measure corrected (Appendix A), the agreement of *k/t*, *sh/ch* and *-ey/-dy* together is +0.132 between adjacent words (95% interval by bifolium 0.111–0.153) and +0.091 at two and three words (0.069–0.113); it halves in about three words. It is almost the same between very different words (+0.09) as between similar ones (+0.13), unlike the copying generator of Timm and Schinner (2020); it is not an effect of position in the line; it is separate for each choice (agreement between different choices –0.01); and it is the same in hands 1, 2 and 3 (+0.12 to +0.15), in languages A and B and in the three transliterations. Agreement of this size exists in languages as grammatical agreement (Italian *-o/-a*, Latin *-us/-a*); the short state alone therefore does not set the Voynich apart from a language with agreement. Its comparison with scribes is in Section 6.2. Local clustering of similar forms was noted by Timm (2014) and Schinner (2007), and a switch of the vowel after *ch/sh* that holds for whole folios by Parisel (2026b); we have not found the short, word-independent state described before.

A slow state. Besides the short state there is a slow component that crosses the line break: words of consecutive lines agree (+0.05), at two lines less (+0.03), at 7–12 lines not at all. It is not a trend from the top to the bottom of the page, and it is normal in real scribes (Section 6.2).

Drift along the line. Inside the line, for the same word and leaving out the first and last word, the marked variants *qo*, *k*, *sh* and *-ey* become rarer towards the right: from –0.020 to –0.034 per ten glyphs, that is about 8–13 percentage points over a line of median length (39 glyphs). The drift is clearest for *sh/ch* ($z = -6.3$ by bifolium); for the other three choices it has the same direction but each one alone does not pass the strictest multiplicity correction. It is steeper in the first half of the line, present in every hand and about three times stronger in language B. The rightward indices of Feaster (2021, 2023) (*qo*- further left than *o*-, *Sh* further left than *ch*) describe the same tendency; we measure it for the same word and compare it with scribes. Next to drawings the marked forms become rarer, and the *q* drops in the word right after a drawing (–33 points); without checking the images we cannot tell whether this is writing or transcription.

5.6 Repetition

One adjacent pair of words in a hundred is the same word repeated (*chol chol*, *qokeedy qokeedy*), and 3.9% are nearly identical (one glyph different), as reported by Boxer (2023) and Bown and Lindemann (2021). In version 1 we read this as repetition beyond chance. The observed/expected ratio corrects that reading (e3c88):

- against the composition of each line (words shuffled within the line), the Voynich repeats the previous word as often as chance (ratio 1.04), whereas every comparison text avoids it: languages 0.2–0.92 (the highest is Sanskrit), the Nordic scribes 0.02–0.07, the German manuscripts at most 0.79, the Copiale copyist 0;
- against the whole vocabulary ($\sum p^2$), the ratio is 2.85: repeated words gather in the same line (Sanskrit reaches 2.86).

The property of the Voynich is therefore that immediate repetition is *not avoided* and that repeated words cluster within lines; nearly identical neighbours are not distinctive (ratio 1.08, with four languages higher). Within the line, repetition is strong between nearby words and fades after six or seven words, whereas in languages it grows with distance (median ratio far/near 2.86); two languages of 20 fall below the Voynich, both with unstable negative ratios. When a repeated word changes its start or end, the change follows the junction rule with its neighbour (*okeey qokeey*).

5.7 Paragraphs, pages and bifolia

- **The paragraph opens with a large sign:** 83% of paragraph-initial words begin with a gallows, against 9% elsewhere, and *p* dominates (54%) (Currier, 1976; D’Imperio, 1978; Feaster, 2023; Stafford, 2023). The opening gallows is three times less tied to its word than usual: it is largely a mark of position.
- **First and last lines have their own register:** longer words and more *p*, *f*, *sh* and final *-y* in the first line; fewer *qo-* and gallows, more *ch-*, *s-*, *y-* in the last.
- ***-ey* grows down the page:** for the same position in the line, the share of *-ey* among words in *-ey/-dy* rises by 12 percentage points from top to bottom of the page (95% interval 7–18) and by 10 points from the first to the last line of a paragraph, separately; lines get shorter down the page.
- **Each page has a strong identity,** mainly along an axis from words in *-edy/-eey* to words in *-aiin/-ain*: pages depart from the profile of their section 2.8 times more than chance.
- **The bifolium is a unit of identity:** the identity of a page is shared by the other side of the same leaf and by the conjugate leaf of the same bifolium, which in the bound book can lie far away, and not by the facing page ($z = 6.6$). The two halves of a bifolium share rare words and copy from each other. All 48 bifolia with enough text are entirely in one language, and 94% in one hand. Hands and languages are recorded page by page in the files, and Davis (2025) had noted that in the herbal section the hands follow the bifolia; the shared rare words and the copying between conjugate leaves are what we add. Davis prefers, tentatively, quires written by one scribe and later bound out of order to scribes working on loose bifolia; our data do not separate the two.
- **Languages A and B differ in vocabulary, not in glyphs:** the best substitution of glyphs brings A closer to B by only 1% of the gap, while the same search recovers a random key every time. B is not A enciphered with another key.
- Consecutive pages share more words than two pages of the same section, but mostly because they are the two sides of one leaf; this does not show that the present order of the leaves is the order of writing, and the binding is known not to be original (Davis, 2025).

6. Comparisons

6.1 Natural and constructed languages

Table 1 and Figure 1 compare the Voynich with the 24 natural languages, one value per language (e3c84). Version 1 compared it with “71 languages”, which were in fact 71 texts, some of them in constructed languages, and two of the reported maxima came from Toki Pona. Summarised per natural language, the statements of version 1 that the Voynich lies beyond all languages hold for the attested wrong cuts, the mirroring of within-word rules, the frequency–form correlation

and the two repetition rates; spaces and form filling are exceeded only by Chinese in pinyin, whose words are syllables; the junction is exceeded by Sanskrit and Tagalog; margin avoidance by Portuguese. Section 5.1 shows that four of these properties follow from word form; the comparison says how unusual the word form is, not that the text is a chain.

Table 1: The Voynich against the 24 natural languages of Gaskell and Bown (2023), one value per language (median of its texts; first 10,000 words), and against the constructed languages. “Beyond” counts the languages further than the Voynich in its own direction; the sensitivity columns use language and period, and the text of each language closest to the Voynich. Experiment e3c84.

measure	Voynich	languages: median (range)	languages beyond: per language / period / closest text	constructed beyond
F1 of spaces from the two flanking glyphs	0.862	0.644 (0.520–0.951)	1 / 1 / 1	none
form filling	0.434	0.046 (0.016–0.465)	1 / 1 / 1	none
attested wrong cuts	0.499	0.065 (0.037–0.236)	0 / 0 / 0	none
mirroring of within-word rules (ρ)	0.470	0.030 (−0.252–0.371)	0 / 0 / 0	none
frequency follows form (ρ)	0.583	0.087 (−0.018–0.260)	0 / 0 / 0	none
junction (bits)	0.191	0.097 (0.034–0.220)	2 / 3 / 4	none
identical neighbours	1.01%	0.13% (0.01–0.40%)	0 / 0 / 0	none
near-identical neighbours	4.13%	0.25% (0.06–1.86%)	0 / 0 / 0	none
left margin (observed/expected)	0.51	0.97 (0.42–1.35)	1 of 23	Lojban
repetition decline within the line	0.12	2.86 (−3.89–12.85)	2 of 20	Lojban, Toki Pona

6.2 Seventy-nine medieval manuscripts and a cipher copyist

If the rules of Section 5 were ordinary scribal habits, real scribes copying real texts would have them. We measured the same properties, with the same code, in 79 manuscripts transcribed at the level of letter forms with their original lines and pages: six Nordic manuscripts from about 1200 to 1550 (Menota) and 73 German manuscripts from 1350 to 1500, the area and century of the Voynich, with lines of five to eight words (Wegera et al., 2021). A variant choice was measured only if it occurs at least 300 times in its minority form among words that show both forms; this rule excludes the choices that are fixed by position in the word (for example a long *s* that never appears at the end). Table 2 gives the results and Figure 2 the intervals choice by choice.

What it means. Junction rules, left-margin avoidance and the non-avoidance of immediate repetition have no counterpart in these manuscripts; line-edge forms exist in scribes but are four to eight times weaker; the drift is smaller in every choice (at most 0.008 against 0.020–0.034 per ten glyphs) and absent with adequate power in 20 of 29. The short state is the property for which the comparison is weakest: with the data available it can be called absent in only half of the choices, and the Swedish scribe of Holm A 10 shows, in the choice between round and straight *r*, an agreement half as large as the Voynich’s that holds between different words too (e3c91, uncertain). The slow state is normal in handwriting. The Copiale copyist does the opposite of the Voynich: having used one symbol for a letter, he tends to use another in the next word (agreement −0.18, down to −0.48 for *u*), as one would expect from someone rotating

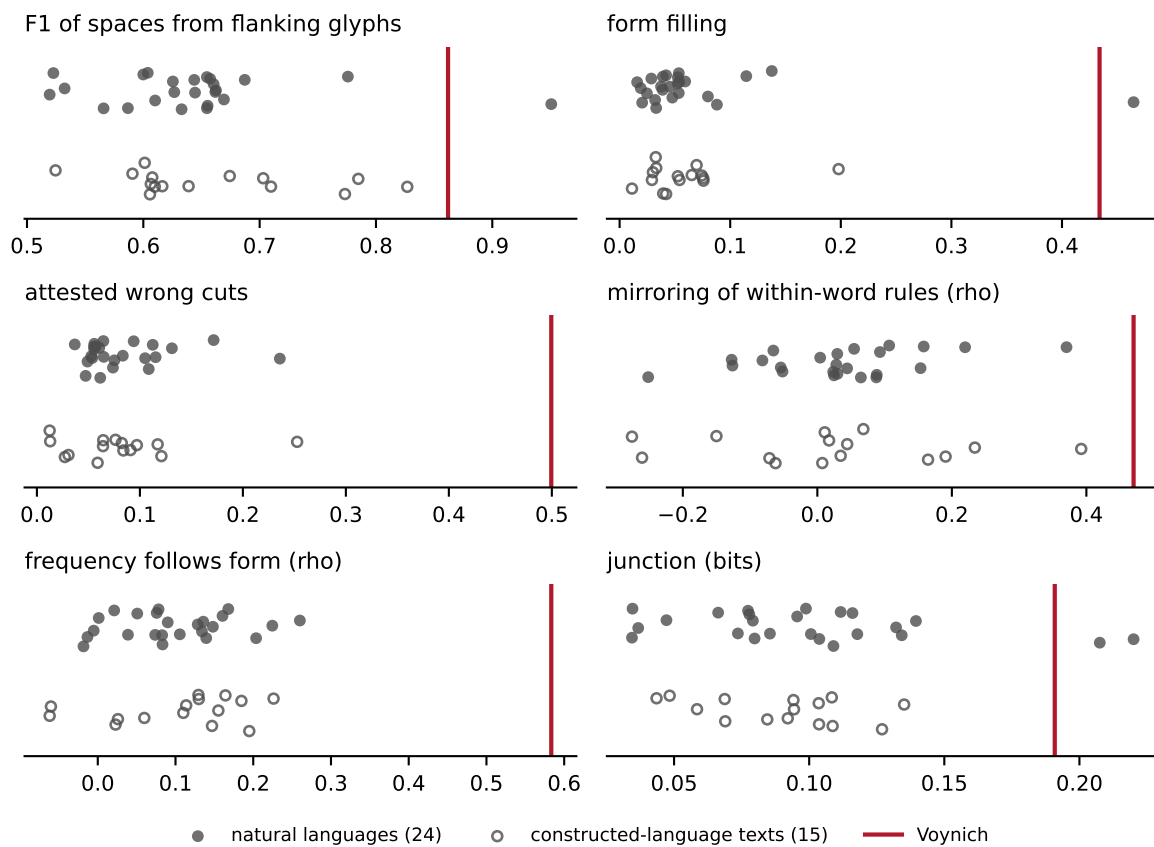


Figure 1: The Voynich (red line) among the 24 natural languages (dots, one per language) and the constructed languages (open circles) for six measures. Data: e3c84.

Table 2: The scribe battery: properties of the Voynich measured identically in 79 medieval manuscripts and in the Copiale. “Absent with power” means that the upper end of the 95% interval lies below half the Voynich value (e3c90).

property	Voynich	6 Nordic manuscripts	73 German manuscripts	Copiale copyist	reading
junction rule (effect / uncertainty of the choice)	0.058	2 of 11 choices at 0.013–0.015, others zero	2 of 13 at 0.001–0.005, others zero	—	about 4 times the strongest Nordic choice, 12 times the strongest German one
left margin: avoiding the start of the line above	5 starts avoided	none in 72 tests	no manuscript with 3 or more starts avoided; 3 of 72 with one, as expected by chance	—	not found in the manuscripts
line-edge forms (gain of the most typical form)	end $-m$ +0.151; start $s-$ +0.098	+0.016 to +0.025 (abbreviations at line end, $v-$ at start)	+0.019 (line-break hyphen); +0.012	—	same kind, 4–8 times stronger in the Voynich
immediate repetition, observed/expected within the line	1.04	0.02–0.07	at most 0.79	0	avoided by every scribe, not by the Voynich
drift along the line (per 10 glyphs)	−0.020 to −0.034	at most 0.0075	at most 0.008, in no common direction	—	no choice drifts as much; absent with power in 20 of 29 choices
short state of choices, at 2–3 words	+0.09	absent with power in 14 of 29 choices; not measurable in 13; present in 2: AM 302 fol f/s (+0.08, from copying similar words) and Holm A 10 round/straight r (+0.05, also between different words; uncertain)	rotates homophones: −0.18 between adjacent words		weaker or absent in scribes; not established as unique
slow state between consecutive lines	+0.050	6 of 17 choices, 5 as large as the Voynich	—	—	normal in handwriting

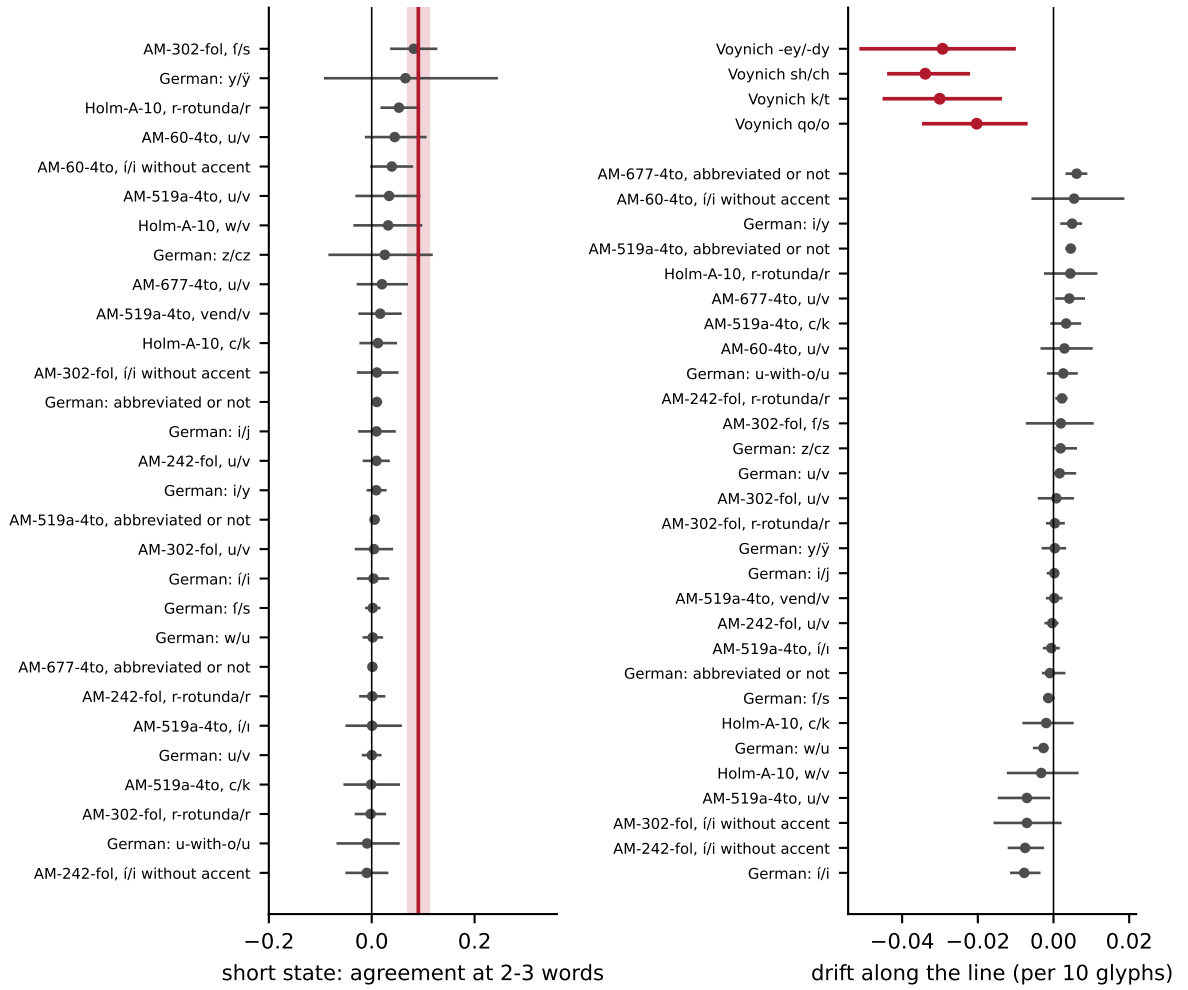


Figure 2: Short state (agreement at two and three words, left) and drift along the line (right) in each choice of the 79 manuscripts, with 95% intervals, against the Voynich (red band: its 95% interval by bifolium). Choices of the German manuscripts are measured on the 73 manuscripts together. Data: e3c50, e3c58, e3c62, e3c77, e3c86, e3c90.

homophones. Fifteenth-century German herbals, recipe books and astrology texts written by hand repeat the adjacent word 7 to 30 times less than the Voynich, so repetition does not come from genre.

These conclusions concern Old Norse and Early New High German manuscripts. Latin or Italian manuscripts of the fifteenth century transcribed at the level of letter forms, the closest in place and kind to the Voynich, are not in the battery.

6.3 Ciphers, readings and generators

Every negative result in this subsection comes from a test that recovers the answer when it is there (positive control).

Encodings of real texts. We encoded real texts (Latin, Italian, the Bible, technical texts) and measured whether they acquire the properties of the Voynich. A word-for-word code takes the letters and the vocabulary of the Voynich but not its repetitions, page homogeneity or junction rules. Verbose ciphers in which each letter becomes a group of two or three glyphs (450 variants) do not reach the predictability of the Voynich with words of its length. The Naibbe cipher, in which each letter or pair of letters becomes a whole word chosen from homophonic tables, does: its predictability and word length are close to the Voynich’s, and Kinnison (2026) shows that it reproduces the positional entropy profile. It does not reproduce repetitions, unique words, page homogeneity, the junction (almost zero) or the line properties. Abbreviations, null letters and transpositions move away from the Voynich; encoded lists, prayers and litanies take on its grain but not its anomalies.

Substitution readings. Homophonic substitution with spaces in 14 languages recovers 69–98% of the words of the controls and at most 36% of Voynich “words”, mostly the same three. A solver without spaces (simulated annealing on a five-gram model of Latin trained on nine million letters, recovering 98–100% of the keys of its positive controls) never reliably beats the negative controls on the Voynich, whatever the glyph inventory, the grouping or the reading direction. Further tests (85 languages with spaces; 16 vernaculars, historical languages and abbreviated Latin; a mixed nomenclator; searches guided by plant names and months; other reading orders; pieces of words as symbols) gave no reading; two formally positive outcomes proved to be artefacts. The order of glyphs inside words is language-like, not anagram-like (compare Hauer and Kondrak, 2016).

Published readings. Bax (2014): his recurring words do not behave as names (*shor*, read as “black”, occurs 96 times on 67 pages of all sections), and against 10,000 random keys his key does not beat chance on the pages not used to build it ($p = 0.14$). Vatne (2021): the reading relies on a transcription of its own; for the 28 of 84 entries that match the reference transliteration, the claim that these words occur nowhere else is false (12 occur on other pages, as often as other paragraph-initial words); the other two thirds could not be tested. Cheshire (2019): the key reads more Romance words than chance, but any key of the same form adapted to the manuscript does better, also on pages not used to adapt it (20 keys of 20). Schechter (2026): the program, re-implemented, gives the published figures, but a “glossary” of the most frequent words without any meaning covers as much, the glossary “reads” 67–70% of a text without a message, and the decoded word order is not Latin. Gatta (2026): the signal is equally present in a text without a message, and a mapping searched for on purpose does better; the current version of the work itself concludes that the text does not behave like Hebrew. Applying

a reading to a text generated without content is simple, and we propose it as a minimum test for any announced decipherment.

Historical mechanisms. We looked for the signature that some encryption systems would leave, after checking on texts enciphered on purpose that each test sees it. Several of these mechanisms are later than the vellum, so these are tests of a kind of mechanism, not historical hypotheses. Homophones chosen by the copyist, as in the Copiale, would leave a negative agreement between nearby words; the Voynich has a positive one. No frequent word or sign behaves like the index letters of Alberti’s cipher disk (c. 1466; Alberti, c. 1466). A message in the binary choices, as in Bacon’s biliteral cipher (Bacon, 1623), restarting at each page or line, would be seen with overwhelming evidence (z up to 70 in the controls) and is absent; a continuous message is out of reach of the test. The word tables of Trithemius (1518) leave no rhythm of two to six words. An autokey restarting at each line is the only mechanism tested that closes the line like the Voynich, but it destroys repetition, inflates the vocabulary (4,500–6,400 distinct words per 10,000, against 2,900) and leaves no margin avoidance.

Constructed languages, gibberish and generators. No constructed language has the junction rules, the line properties or an agreement between adjacent word forms like the Voynich’s. The volunteers’ gibberish has no junction (0.014), no line-edge forms, no copying from the line above and no margin avoidance (it repeats line starts). Among the published generators we measured, Naibbe has neither junction nor line properties; the generator of Timm and Schinner (2020) copies from too far above, has no junction closed within the line, no word-independent state, no margin avoidance and no drift; U2 and U3 (Whitehatnetizen, 2026) do not combine the junction with a closed line. We have not run *voynich-fingerprint* (Sachak, 2026), whose 44 statistics, reproduced on held-out folios, include a cross-word junction computed without line breaks but no line-final forms, no vertical relation and no margin; our statement is therefore limited to the generators measured.

7. From measurements to a scorecard

A generator needs a target that can be checked. The measurements were turned into four instruments, all computed on a whole generated book and on the real manuscript in the same way.

The scorecard. Eighteen properties (Table 3), each with a pass band around the Voynich value. Seventeen come from the early experiments of the project and one (the line edges) from a later one; the bands were chosen after seeing the results of those experiments, so they are a specification, not a test. The book is cut into pseudo-pages of 29 lines. Measured in exactly this way, the real manuscript passes 17 of the 18: the band of the “gradient” (similarity at six lines relative to similarity within the line) was set from a different page division and the manuscript itself scores 0.68 against a band of 0.70–0.97. We therefore count 17 attainable properties.

Eight extra properties. Added later from the research results (e293): rare words per page, types per token and unique words within the page, the spread of word lengths, the register of the first lines of paragraphs, the spelling choices that vary from line to line (at least 10 of 12 classes with excess variance), the agreement of word endings, and the share of word pairs seen elsewhere in the text. The manuscript passes all eight.

The line gate. Five conditions that must hold together: a line rarely starts with the first glyph of the line above (ratio at most 0.7; Voynich 0.53); the junction across the line break is below a tenth of the junction inside the line; similarity at distance two is at least that at distance one (Voynich 1.05); at least three of the five spelling choices vary from line to line ($z > 3$); and the correlation between consecutive lines is 0.207 ± 0.07 .

Two judges. A judge is a classifier trained to tell real pages from generated ones, retrained from scratch on every generated book; its cross-validated area under the ROC curve (AUC) is a classifier two-sample test (Lopez-Paz and Oquab, 2017). Each real page and the generated page with the same folio name are kept in the same fold (10 folds, repeated with five seeds), on the 202 pages with at least 40 legible words. The first judge (e231) is an L2-regularised logistic regression on 209 page features: frequencies of the 30 most frequent glyphs and of the 150 most frequent glyph pairs inside words; word length; types per token and unique words; first and last glyphs of lines; similarity of neighbouring words and attested joins; and vertical similarity. The second (e266) adds 19 features: pairs seen elsewhere, identical neighbours, similarity at distance two, the length and initials of first and last words of lines, galls and word length in the first lines of paragraphs, and the divergence of a page’s glyphs from the book’s and between the halves of the page. No feature tests whether a word belongs to the Voynich vocabulary. An AUC of 0.5 means that the judge is guessing and 1 that it never errs; the project set in advance 0.6 or less as “hard to tell apart” and 0.7 or more as “distinguishable”. Controls: Currier’s languages A and B are told apart with AUC 1.000; random labels give 0.53 and 0.52; two earlier generators of the project give 0.97/0.99 and 0.87/0.94.

A ladder of controls. To read an AUC, we degraded the real manuscript one level at a time and judged it (e400; Table 4). Shuffling the lines within each page costs nothing to the first judge and brings the second to 0.60; shuffling the words within each line already gives 0.98–0.99; drawing the words from other pages of the same section gives 1.00. The second judge’s floor for any text without the real line order is thus about 0.60.

What the instruments leave out. Some rules of Section 5 enter the instruments only indirectly (the drift along the line, the short state of choices, the bifolium as a unit) or not at all (closure at the gap of a drawing, the labels). The instruments were built by the project and the generator was tuned against them; an independently built judge has not been tried (Section 11).

Table 3: The scorecard: 18 properties measured on a whole book cut into pseudo-pages of 29 lines, with the pass band and the value of the real manuscript. Property 7 is not attainable (the manuscript fails its band). Values of the generator are in Table 6.

#	property (measure)	passes if	Voynich
1	conditional entropy of the next glyph, h_2	± 0.15	2.236
2	predictability of the space	± 0.08	0.664
3	share of words occurring once (at 34,000 words)	± 0.08	0.678
4	types per token, whole text	± 0.04	0.210
5	identical neighbours, relative to the line’s composition	± 0.2	1.009
6	homogeneity: similarity of words within the line	± 0.008	0.0385
7	gradient: similarity at six lines / within the line	0.70–0.97	0.683 (fails)
8	junction: last glyph $\rightarrow$ first glyph	± 0.05	0.188
9	attested joins of neighbours, relative to chance	1.5–2.5	1.97
10	flat curve of once-occurring words (1,000 vs 34,000 words)	≤ 0.10	0.057
11	vocabulary turnover between near and far blocks	≥ 0.05	0.106
12	page profile (ratio and share)	0.8–1.25 and 0.5–1.5 $\times$	1.06; 0.061
13	autocorrelation of neighbouring word lengths	≥ 0.08	0.150
14	Zipf slope	± 0.10	–1.041
15	word shape: glyphs at the first, second, second-last and last position	≤ 1.0	0
16	vertical similarity: same position in the line above	≥ 1.015	1.028
17	formulas: two- and three-word sequences recurring at least ten pages apart	0.5–1.5 $\times$	6.39
18	line edges: divergence of line-initial and line-final glyphs	0.5–2 $\times$	23; 55

Table 4: The ladder of controls: the real manuscript, with its true layout, degraded one level at a time (means of four seeds; e400).

text	AUC judge 1	AUC judge 2	scorecard (of 18)
the manuscript as it is	0.500	0.500	17.0
lines shuffled within each page	0.481	0.595	17.0
words shuffled within each line	0.984	0.990	11.5
words shuffled within the page	0.989	0.998	11.2
words drawn from the other pages of the same section and language	0.998	1.000	6.2
words drawn from the other pages of the book	1.000	1.000	5.0
the manuscript with a message forced into its five spelling choices	0.622	0.630	16.0

8. The voynichizer

The voynichizer (Caniatti, 2026b) takes any text in UTF-8 and a key, and writes a book with the 207 folio names, sections and Currier languages of the manuscript, about 4,050–4,260 lines in EVA transliteration. With the key the text comes back exactly; with a wrong key it is rejected. This section describes version 21, the published one. Figure 3 gives the overall design and Figure 4 an example.

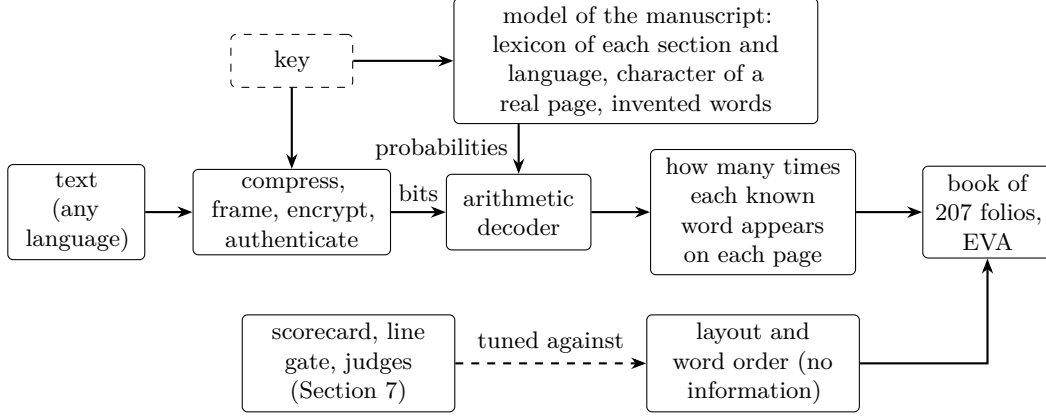


Figure 3: Design of the voynichizer. The encrypted bits only decide the word counts of each page, through an arithmetic decoder that samples them from the model; everything else comes from the key and the manuscript’s statistics. Reading runs the arithmetic encoder on the observed counts.

8.1 Design principle

Feeding uniformly random bits to an arithmetic decoder produces an exact sample of the decoder’s probability model, up to the quantisation of the probabilities. If the bits are the ciphertext of a message under a good cipher, they are indistinguishable from random bits, and the book is a sample of the model whether or not it carries a message: security relative to the model in the sense of Cachin (1998), by the method of Ziegler et al. (2019). Two consequences follow. The model, not the message, decides how close the book comes to the manuscript; and the capacity is fixed by the entropy of the part of the model that carries the bits, and cannot be raised without changing the statistics.

8.2 From text to bits

The text is compressed with zlib and framed as *length* (4 bytes) || *tag* (8 bytes) || *compressed text*, where the tag is a truncated HMAC-SHA256. The frame is XORed with a SHAKE-256 keystream; after it, the same keystream continues as filler, so that a book without a message is driven by the keystream alone. The master key is derived with *sCrypt* ($n = 2^{15}$, $r = 8$, $p = 1$) from the passphrase; every random choice of the program has its own generator seeded from an HMAC of the master key, the use and the page. For example, the opening of Book XVII of Isidore’s *Etymologiae* (11,207 bytes) compresses to 4,776 bytes and becomes 38,304 framed bits.

8.3 The model of the book

Layout. For each page type (section and Currier language), the program keeps the number of lines per page, the page width (median words per line) and paragraph lengths, and for the

whole book the ratio of words in a line to the page width by line role (first, middle, last, only line of a paragraph). It draws a page from these statistics, so no generated page copies the layout of a real one; books come out about 5% longer than the manuscript.

Known and invented words. About 86% of the manuscript’s running words are *known* words (2,246 types that occur at least twice); the rest occur once. In each of eight place classes (first line of a paragraph or not, times first, second, middle or last word of the line) a place receives an invented word with the probability observed for once-occurring words in that class in the manuscript, adjusted by the page. Invented words are generated by a glyph model with four-glyph contexts learned on the 4,776 once-occurring words, with lengths drawn by place class; a quarter of the long ones are joins of two known words of the page whose junction is probable. An invented word is never a Voynich word, so it cannot be confused with a known one when reading.

The lexicon of a page and its “character”. The known-word lexicon of a page is made of the occurrences of known words on the other pages of the same section and language (never the page itself). The key chooses another real page of the same type, whose glyph frequencies give the page its “character”: each word’s weight is its count raised to $\gamma = 0.976$, times $\exp(\kappa \sum_g n_g(w) \delta_g)$, where δ_g is the smoothed log-ratio of glyph g between the chosen page and its section and $\kappa = 0.644$.

8.4 The channel: word counts

The bits decide only how many times each known word appears on each page. For a page with n known-word places and ordered integer weights $w_1 \geq \dots \geq w_m$ (total W), the counts are one multinomial draw, written as a chain of binomials: $c_1 \sim \text{Bin}(n, w_1/W)$, $c_2 \sim \text{Bin}(n - c_1, w_2/(W - w_1))$, and so on. Each binomial is a run of binary decisions (“stop at j copies?”, with probability $P(X = j)/P(X \geq j)$ quantised to 16 bits), and each decision is a symbol of a 32-bit binary arithmetic coder. Writing runs the coder’s *decoder* on the ciphertext bits, so the decisions are sampled; reading runs its *encoder* on the observed decisions and recovers the bits, then XORs them, checks length and tag and decompresses. The bits are consumed in page order; the Isidore text fills about 130 of the 207 pages and the filler the rest.

Capacity. A v21 book carries 81,100–85,900 bits (mean 83,400 over 24 keys), about 2.7 bits per known word, roughly 20,000 characters of ordinary prose after compression. The first versions hid the bits in the five spelling choices (*ch/sh*, *k/t*, *-l/-r*, *qo-/o-*, *-dy/-ey*), with about 0.8 bits per choice and 46,000 bits per book; moving the channel into the word counts nearly doubled the capacity and made the books harder to tell apart (Table 5).

8.5 Arrangement: word order carries no information

Once the multiset of words of a page is fixed, its order is chosen by a Metropolis sampler (60 proposed swaps per word, temperature 1) on a score with thirteen terms learned or tuned on the manuscript: the affinity of each word for its place class (a logistic model on the word’s first and last glyphs, length, prefix and ending); links between neighbouring words (ending–prefix and prefix–prefix, last glyph–first glyph, joins that form a word of the book, pairs seen on other pages, identical neighbours); similarity with the word at the same position in the line above and with the word two places away; avoidance of a line starting with the same glyph as the line above; agreement of the five spelling choices within a line and between consecutive lines;

a difference between the two halves of the page; and line widths. None of these choices affects reading: moving words within a page changes nothing; adding, removing or changing a known word breaks the reading, and there is no error correction.

8.6 Security, as far as it goes

The building blocks are standard, but the whole has not been reviewed by a cryptographer. The script salt is fixed, so a dictionary of guessed passphrases can be prepared once for all books; the keystream depends only on the key, so two books written with the same key share it and a different key should be used for each message. A long random passphrase (six random words or twelve random characters) is needed. The program hides *that there is a message*, not that the program was used: almost every word of a book is a Voynich word, and a book is recognisable as the program’s output.

8.7 Development

Table 5 summarises the versions that changed the method. Every version was judged on the same instruments; from version 14 on, a version was adopted only after confirmation on 12 keys never used for development.

Table 5: Main versions of the voynichizer (AUC of the two judges; “3 keys” are early benchmarks, optimistic compared with later replications).

version	what changed	AUC judge 1 / judge 2
v0	body from an earlier generator; message forced into the five spelling choices	0.958 / 0.980
v1	arithmetic coding over a model of the spelling choices; about 46,000 bits	0.889 / 0.957
v8	generator “by pieces”: page lexicon, borrowed character, invented words, learned arrangement	0.604 / 0.715 (3 keys)
v10	message moved into the word counts	0.572 / 0.690 (12 keys)
v14	joins accepted only with a probable junction	0.575 / 0.600 (24 keys)
v15–v17	authenticated encryption (scrypt, SHAKE-256, HMAC); statistical layout; first public release	0.559 / 0.597 (24 keys)
v21	correction of an initialisation defect in the arrangement (found by reading the public code)	0.554 / 0.601 (24 keys)

f1r, the manuscript (first 12 lines, ZL transliteration)

Դապցն ցլի ախ ալ ա՛լաւո ձօս ձօնց ժիւնն ց լի օր ձօսծց
 ճօնց ժիւալ օր ց լի ալ ա՛լաւո ձալ աճն ժիւալ ժիւալ ծաւ
 ճցաւ ձիւլ ց օր ցլի աւո ձօծ ժիւոալց ժիւոոն ծաւաւո ճց
 ճօւո օ՞լլոց օ՞լլոն լօսօ՞լլց ժիւալ ծաւո օ՞լլաւ օր օ՞լլաւ
 ճալ ց աճալ ժիւաւո ժիւալ ժիւաւո
 ցծաւաճց
 օծալ ճց ձօս ժիւոց օցծալ ձիւ ճ ժիւոաւո ձօծալց
 ցձոց ձօծց օ՞լլաոց օ՞լլաօս աօժիւց օճն աց ծաւ աօր լի օճ
 ծաւո ձօն ժիւօս ձօծց
 ծաւ օճ լիւծց
 ցծաւ ժիւոաւո օսն ժիւոց ցլիւո ձօնց ժիւծալ ճն
 օ՞լլ ձօ լի ձօց օ՞լլաւո օ՞լլոս օ՞լլաւ ձօծաւ ճժիւց ծաւ

f1r, a book generated by version 21 (first 12 lines)

օաժօրօւաւո ձիւալ օ՞լլաոց օ՞լլալ ժիւոց օ՞լլաոց ալ ձիւ աօր
 ճօւո աւ աճոն օսսց օւոն ցլի ծաւ ծալ աճիւ աց ծաւ
 ճաւ օսաճն ալ ձիւոց օսլիալ աճ ալ օսաճց օ՞լլաւ օսլի օս ցլի
 ցլիաւ ցլի աճց աօս լի ալ աճիւ աւ օսլի օս աւ օլլոց օ՞լլաւ
 ցլօօս ցլի աց օլաւ ալօծոց ճաճց օ՞լլաւ օ՞լլալ
 օսլի ց օս օ՞լլաօս օ՞լլաւ աճալ լ 4օ լի օո լ ձիւց օսլի ց 4օլլաւ օր օսաճց 4օլլալ
 օլլալ լց
 ճաւ ձիւոց 4օլլալ ձիւոց ճլի ալ օսլի օոց 4օոճի օոց 4օլլալ
 ցլի աց լի աճց ծալ աժօլլ ց ճօս աճիւ աօր ծաճոց աաւ աճց ցլի օս 4օլլալ
 ճօլաւ ծալ
 4օ ճաճց 4օլլալ ց 4օլլաւ օ՞լլոց ճլի օս 4օլլալ օս ց ցլի աճոց
 օր աաւ օ՞լլաւ օրլի օսց
 լիոց լի օոց աօս ձիւոց օ՞լլաճ օ՞լլ օ լի օոց օ՞լլո 4օլլալ օ՞լլոց ցլի աճոց
 ճաւ օսաւ օ՞լլաւ աօլլ ց ձիւ օս օլլոց աօսլի օս ց օ՞լլաւ աճ օր ձիւ օս

Figure 4: The first twelve lines of folio 1r in the manuscript (top) and in a book generated by version 21 (bottom), both set in the voynichizer’s font. The generated book hides the sentence “This book was written by the voynichizer. Its words hide this sentence, compressed and encrypted; without the key it reads like the Voynich manuscript and says nothing.” It can be read back with the public program and the key `white paper figure one, public example key`. The message uses 1,144 of the book’s 83,480 bits.

9. Evaluation

Protocol. Version 21 was run with 24 keys on the same text (the opening of Isidore’s *Etymologiae* XVII, 38,304 bits). Keys 1–12 had also been used during development to choose between variants; keys 13–24 never were. Each book was judged by the two judges, the scorecard, the eight extra properties and the line gate, read back with its key and with a wrong key.

Table 6: Version 21 on 24 keys: the instruments of Section 7 (mean $\pm$ standard error over keys). Lower part: each scorecard property, Voynich value, mean of the generated books, and the number of keys whose book fails it.

	keys 1–12	keys 13–24 (fresh)	all 24
AUC judge 1	0.554 ± 0.006	0.554 ± 0.005	0.554 ± 0.004 (0.517–0.580)
AUC judge 2	0.596 ± 0.008	0.606 ± 0.006	0.601 ± 0.005 (0.551–0.632)
keys with judge 2 above 0.60	6	8	14
scorecard (of 17 attainable)	16.0	15.9	16.0
extra properties (of 8)	5.5	5.2	5.4
line gate passed	12 of 12	12 of 12	24 of 24
exact read-back; wrong key rejected	12; 12	12; 12	24; 24
capacity (bits)			81,100–85,900

#	scorecard property	Voynich	generated (mean)	keys failing
1	h_2	2.236	2.284	0
2	predictability of the space	0.664	0.639	0
3	words occurring once	0.678	0.738	0
4	types per token	0.210	0.205	0
5	identical neighbours relative to the line	1.009	0.943	0
6	homogeneity within the line	0.0385	0.0414	1
7	gradient (not attainable)	0.683	0.62	24
8	junction	0.188	0.180	0
9	attested joins relative to chance	1.97	2.13	0
10	flat curve of once-occurring words	0.057	−0.024	0
11	vocabulary turnover	0.106	0.095	0
12	page profile (ratio)	1.06	0.66	24
13	autocorrelation of word lengths	0.150	0.142	0
14	Zipf slope	−1.041	−1.005	0
15	word shape	0	0.040	0
16	vertical similarity	1.028	1.035	0
17	formulas	6.39	4.58	0
18	line edges (start; end)	23; 55	23.6; 54.5	0

How close to the manuscript. The first judge separates generated from real pages with AUC 0.55, the second with 0.60 (Table 6, Figure 5); on the fresh keys the second judge gives 0.606. For the second judge this is the level of the real manuscript with its lines shuffled within each page (0.595), and far from the manuscript with its words shuffled within each line (0.99). The group of features that the second judge uses best is the register of the first lines of paragraphs (AUC 0.62 on that group alone). The books pass 16 of the 17 attainable scorecard properties: they always miss the page profile, a measure of how much each page departs from its section, and once the homogeneity within the line. Of the extra properties they usually miss the spelling choices that vary from line to line (23 of 24 keys), the register of the first lines (18

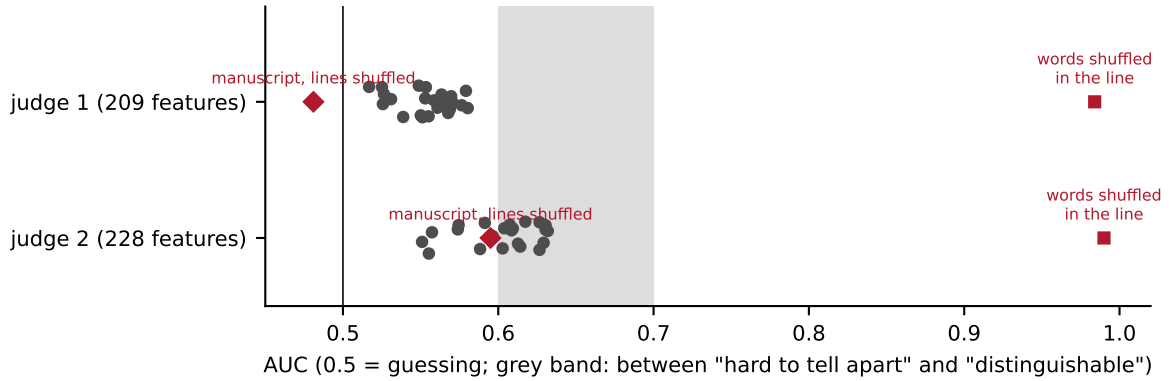


Figure 5: AUC of the two judges for the books of the 24 keys (dots), against the real manuscript degraded on purpose (red): with its lines shuffled within each page, and with its words shuffled within each line. Data: e409b (version 21) and e400.

of 24) and the spread of word lengths (20 of 24). Lines are too irregular in width: 11.6% are wider than 1.25 times the page median, against 6.0% in the manuscript. About 0.4–0.5% of the word triples of a book also occur in the manuscript.

With and without a message. [e419: with 24 keys, books with the Isidore text and books with filler only; differences of the two judges against the manuscript, and a direct classifier between the two kinds of book.]

What the generator does not do. It writes only running paragraph text: no labels, no circular or radial text, and no images (the website places the real drawings next to the generated folios). It does not model hands, and it generates pages independently, so the bifolium effects of Section 5.7 are absent. It models copying from the line above by the index of the word in the line, whereas the manuscript copies by physical position. Two rules of Section 5 that are not in the instruments, the closure at the gap of a drawing and the drift along the line, were not targeted, and the short state only as agreement of the spelling choices within a line.

10. Discussion

10.1 What the generator shows

The voynichizer is a constructive argument. Its books reproduce most of the measured writing rules of the manuscript, to the point that two classifiers retrained on each book separate its pages from the real ones only slightly better than chance, and every book carries an encrypted text that comes back exactly with the key. Two claims must be kept apart.

- **A book with a message and a book without one** are samples of the same distribution, because the message enters only as encrypted bits, which are indistinguishable from the random filler that drives a book without a message. No test on the text can separate them without the key. This holds by construction, whatever the judge, as long as the cipher is sound; the measurement on version 21 is in Section 9.
- **A generated book and the real manuscript** are not the same distribution. Our judges separate them a little (AUC 0.55 and 0.60), several properties are not reproduced, and a

stronger or independently built judge would probably do better. How close the book comes is a measured fact about our instruments.

Together they mean that statistical regularities of the kind measured here, however strong, cannot decide by themselves whether the Voynich manuscript carries a message: a text written under these rules can carry one, and whether it does cannot be read from its statistics. This does not show that the manuscript carries a message. It shows that the debate between “meaningless” and “enciphered” cannot be settled by the statistics of the text alone (compare Bown and Gaskell, 2023; Gaskell and Bown, 2023; Greshko, 2025; Kinnison, 2026).

10.2 What the measurements exclude and what they leave open

Excluded, with the controls described.

- a natural language written in an ordinary alphabet and read with a simple or homophonic substitution, in the languages tested;
- the published readings we could test;
- the historical encryption mechanisms we simulated, one at a time;
- a text invented by hand without rules, as the volunteers’ gibberish is;
- that the rules of the manuscript are the ordinary habits of the 79 Nordic and German manuscripts we measured (with the reservations on the short state in Section 6.2).

Open. Four scenarios, which our data do not separate:

1. **A language written with a very artificial writing system**, in which the variant forms carry no meaning but follow rules of form and spaces are placed by rules of form. The junction rules are compatible with a phonetically written language: two of the 24 languages, Sanskrit and Tagalog, have a stronger junction.
2. **A code or cipher of an untested kind**, for example with a repertory of words or a lost key book. The voynichizer shows one way in which a message can hide in a text with these rules.
3. **No message**: a text produced by a procedure that can be executed by hand. Generators of this kind reproduce many basic properties, but none of those we measured reproduces the junction rules, the closed line and the margin avoidance together.
4. **A line that is a unit of content** (a verse, an entry, a recipe, a formula). It would explain at once the closed line, the line-edge forms and the register of first and last lines; a herbal printed one verse per line has a closed line too. It is compatible with any of the first three scenarios.

10.3 What would change our mind

- A reading that passes the controls used here: one that works on the real text and not on shuffled text or on text generated without a message, that produces meaningful sentences on pages not used to build it, and that explains the junction rules, the closed line, the copying from the line above and the margin avoidance.
- A medieval manuscript, or a historical cipher, with the same rules; in particular Latin or Italian manuscripts of the fifteenth century, which the battery lacks.

- A simple procedure, executable by hand, that reproduces them all together.

A reading will need an external anchor that regularities cannot give: a text of which the manuscript is a copy, a secure identification of many plants to use as cribs, a key found in an archive.

11. Limitations

- **Preregistration and exploration.** The preregistrations are internal, dated files in a public version-controlled history, without third-party timestamps. The programme was exploratory, and confirmations on data not used to form a hypothesis are few (Section 4).
- **Transliterations.** Everything rests on transliterations that share the line segmentation. Without checking the images we cannot separate what the scribe did from what the transcribers read, for example the *q* that drops after a drawing; for uncertain spaces, the image measurements of Rozanova and Temerev (2026) show that they are physically narrower.
- **Comparison manuscripts.** Old Norse and Early New High German only; no fifteenth-century Latin or Italian manuscript transcribed at the level of letter forms. The files do not mark changes of hand, and the 73 German manuscripts were measured together for the state of the choices, so a single scribe’s state could be diluted. For the short state the power is insufficient in 13 of 29 choices.
- **Comparison languages.** 24 natural languages, with lines made by cutting prose; 10,000 words per text.
- **Historical ciphers.** Only one real enciphered manuscript (the Copiale); the DECODE archive of early modern ciphers requires an account and has no reuse licence, and we found no fifteenth-century cipher there with both transcription and key.
- **Gaps.** Three tests could not be run because their measures did not separate the cases on synthetic texts: whether the state of the choices lasts a number of words or a span of writing (glyphs), whether the states of different choices change at the same points, and whether the short state crosses the line break. Whether the state survives the gap of a drawing is undecided for lack of data.
- **The generator.** The judges and the scorecard are the project’s own, and the generator was tuned against them; an independently built judge has not been tried. Keys 1–12 were used during development. The books are not indistinguishable from the manuscript. The encryption has not been reviewed by a cryptographer, and the script salt is fixed.
- **Effect sizes.** Several of the properties are small in absolute terms (the short state is about 13 percentage points of agreement between adjacent words); the look-ahead and the drift of single choices other than *sh/ch* do not survive the strictest multiplicity correction.

12. Availability

- **Research record** (Caniatti, 2026a): preregistrations (`preregistrazioni/`), programs (`esperimenti/`), results with provenance records (`risultati/`), the laboratory notebook (`QUADERNO.md`, in Italian), the log of all experiments (`white_paper/revisione/registro_esperimenti.csv`) and the sources of this paper with the script that draws its figures. Re-running an experiment gives the same numbers. Copyright-protected corpora are not included and must be downloaded from their sources (Appendix E).

- **The voynichizer** (Caniatti, 2026b): program (MIT licence), the transliteration it uses (CC0), the font, and a website that runs version 21 in the browser. The example book of Figure 4 and its message are in the research repository ([white_paper/en/figure/](https://github.com/AndreottiVIII/voynichizer)).

References

- Leon Battista Alberti. *De componendis cifris*, c. 1466.
- Francis Bacon. *De dignitate et augmentis scientiarum*. 1623.
- Stephen Bax. A proposed partial decoding of the Voynich script, 2014. Self-published PDF, January 2014. Archived copy: <https://web.archive.org/web/20140307232358/http://stephenbax.net:80/wp-content/uploads/2014/01/Voynich-a-provisional-partial-decoding-BAX.pdf>.
- William Ralph Bennett. *Scientific and Engineering Problem-Solving with the Computer*. Prentice-Hall, Englewood Cliffs, 1976.
- Claire Bown and Daniel Gaskell. Enciphered after all? word-level text metrics are compatible with some types of encipherment. In Colin Layfield and John Abela, editors, *Proceedings of the 1st International Conference on the Voynich Manuscript 2022 (VOY2022)*, volume 3313 of *CEUR Workshop Proceedings*, 2023. Paper 6.
- Claire L. Bown and Luke Lindemann. The linguistics of the Voynich manuscript. *Annual Review of Linguistics*, 7:285–308, 2021. doi: 10.1146/annurev-linguistics-011619-030613.
- Alexander Boxer. Fingerprinting gibberish: A quantitative comparison of the Voynich and Sloane MS 3188. In Colin Layfield and John Abela, editors, *Proceedings of the 1st International Conference on the Voynich Manuscript 2022 (VOY2022)*, volume 3313 of *CEUR Workshop Proceedings*, 2023. Paper 1.
- Christian Cachin. An information-theoretic model for steganography. In *Information Hiding, Second International Workshop (IH'98)*, volume 1525 of *Lecture Notes in Computer Science*, pages 306–318, 1998. doi: 10.1007/3-540-49380-8_21.
- Davide Caniatti. Statistical study of the Voynich manuscript: preregistrations, code, results and laboratory notebook, 2026a. <https://github.com/AndreottiVIII/voynich>.
- Davide Caniatti. The voynichizer (voynichizzatore), version 21, 2026b. Software, MIT licence. <https://github.com/AndreottiVIII/voynichizzatore>, commit 4b1383c; web version <https://andreottiviii.github.io/voynichizzatore/>.
- Andrew Caruana, Colin Layfield, and John Abela. An analysis of the relationship between words within the Voynich manuscript. In Colin Layfield and John Abela, editors, *Proceedings of the 1st International Conference on the Voynich Manuscript 2022 (VOY2022)*, volume 3313 of *CEUR Workshop Proceedings*, 2023. Paper 8.
- Gerard Cheshire. The language and writing system of MS408 (Voynich) explained. *Romance Studies*, 37(1):30–67, 2019. doi: 10.1080/02639904.2019.1599566.
- Christos Christodouloupoulos and Mark Steedman. A massively parallel corpus: the Bible in 100 languages. *Language Resources and Evaluation*, 49(2):375–395, 2015. doi: 10.1007/s10579-014-9287-y. First author’s name as printed; usually spelled Christodouloupoulos.
- Prescott H. Currier. Papers on the Voynich manuscript. In Mary E. D’Imperio, editor, *New Research on the Voynich Manuscript: Proceedings of a Seminar, 30 November 1976*. Washington, 1976. Transcription by J. Guy and J. Reeds (1992): https://www.voynich.nu/extra/curr_main.html.
- Lisa Fagin Davis. How many glyphs and how many scribes? digital paleography and the Voynich manuscript. *Manuscript Studies*, 5(1):164–180, 2020. doi: 10.1353/mns.2020.0011.

- Lisa Fagin Davis. Voynich codicology, 2025. Manuscript Road Trip (blog), 19 January 2025. <https://manuscriptroadtrip.wordpress.com/2025/01/19/voynich-codicology/>.
- Mary E. D’Imperio. *The Voynich Manuscript: An Elegant Enigma*. National Security Agency/Central Security Service, Fort Meade, 1978.
- Kevin Farrugia, Colin Layfield, and Lonneke van der Plas. Demystifying the scribes behind the Voynich manuscript using computational linguistic techniques. In Colin Layfield and John Abela, editors, *Proceedings of the 1st International Conference on the Voynich Manuscript 2022 (VOY2022)*, volume 3313 of *CEUR Workshop Proceedings*, 2023. Paper 5.
- Patrick Feaster. Ruminations on the Voynich manuscript, 2020. Griffonage-Dot-Com (blog), 24 August 2020. <https://griffonagedotcom.wordpress.com/2020/08/24/ruminations-on-the-voynich-manuscript/>.
- Patrick Feaster. Rightward and downward in the Voynich manuscript, 2021. Griffonage-Dot-Com (blog), 18 August 2021. <https://griffonagedotcom.wordpress.com/2021/08/18/rightward-and-downward-in-the-voynich-manuscript/>.
- Patrick Feaster. Rightward and downward grapheme distributions in the Voynich manuscript. In Colin Layfield and John Abela, editors, *Proceedings of the 1st International Conference on the Voynich Manuscript 2022 (VOY2022)*, volume 3313 of *CEUR Workshop Proceedings*, 2023. Paper 12.
- Daniel E. Gaskell and Claire L. Bower. Gibberish after all? Voynichese is statistically similar to human-produced samples of meaningless text. In Colin Layfield and John Abela, editors, *Proceedings of the 1st International Conference on the Voynich Manuscript 2022 (VOY2022)*, volume 3313 of *CEUR Workshop Proceedings*, 2023. Paper 4. Data: <https://github.com/danielgaskell/voynich> (publicly released subset; modified MIT licence).
- Antenore Gatta. voynich-toolkit, v0.26.0, 2026. Software, Zenodo, 25 March 2026.
- Michael A. Greshko. The Naibbe cipher: a substitution cipher that encrypts Latin and Italian as Voynich manuscript-like ciphertext. *Cryptologia*, 2025. doi: 10.1080/01611194.2025.2566408. Published online 26 November 2025.
- Bradley Hauer and Grzegorz Kondrak. Decoding anagrammed texts written in an unknown language and script. *Transactions of the Association for Computational Linguistics*, 4:75–86, 2016. doi: 10.1162/tacl_a_00084.
- Jürgen Hermes. Polygraphia III: The cipher that pretends to be an artificial language. In Colin Layfield and John Abela, editors, *Proceedings of the 1st International Conference on the Voynich Manuscript 2022 (VOY2022)*, volume 3313 of *CEUR Workshop Proceedings*, 2023. Paper 7.
- Nicholas J. Hopper, John Langford, and Luis von Ahn. Provably secure steganography. In *Advances in Cryptology – CRYPTO 2002*, volume 2442 of *Lecture Notes in Computer Science*, pages 77–92, 2002. doi: 10.1007/3-540-45708-9_6.
- Gabriel Kaptchuk, Tushar M. Jois, Matthew Green, and Aviel D. Rubin. Meteor: Cryptographically secure steganography for realistic distributions. In *Proceedings of the 2021 ACM SIGSAC Conference on Computer and Communications Security (CCS ’21)*, pages 1529–1548, 2021. doi: 10.1145/3460120.3484550.
- Rod Kinnison. The Voynich manuscript’s positional entropy collapse: comparative evidence from languages and ciphers. *Cryptologia*, 2026. doi: 10.1080/01611194.2026.2697318. Published online 7 September 2026.
- Kevin Knight, Beáta Megyesi, and Christiane Schaefer. The Copiale cipher. In *Proceedings of the 4th Workshop on Building and Using Comparable Corpora (BUCC)*, pages 2–9. Association for Computational Linguistics, 2011a.
- Kevin Knight, Beáta Megyesi, and Christiane Schaefer. The secrets of the Copiale cipher.

- Journal for Research into Freemasonry and Fraternalism*, 2(2):314–324, 2011b. doi: 10.1558/jrff.v2i2.314. Published 2012.
- Luke Lindemann and Claire Bown. Character entropy in modern and historical texts: Comparison metrics for an undeciphered manuscript, 2020. arXiv:2010.14697.
- David Lopez-Paz and Maxime Oquab. Revisiting classifier two-sample tests. In *International Conference on Learning Representations (ICLR)*, 2017. arXiv:1610.06545.
- Vladimír Matlach, Barbora Anna Janečková, and Daniel Dostál. The Voynich manuscript: Symbol roles revisited. *PLOS ONE*, 17(1):e0260948, 2022.
- Menota. Medieval nordic text archive. <https://www.menota.org>. Texts used (CC BY-SA 4.0): AM 519 a 4to, AM 677 4to, AM 60 4to, AM 242 fol, Holm A 10, AM 302 fol.
- C. Parisel. Evidence of layered positional and directional constraints in the Voynich manuscript: Implications for cipher-like structure, 2026a. arXiv:2604.19762.
- C. Parisel. A quantitative confirmation of the Currier language distinction, 2026b. arXiv:2604.25979.
- Shravana Reddy and Kevin Knight. What we know about the Voynich manuscript. In *Proceedings of the 5th ACL-HLT Workshop on Language Technology for Cultural Heritage, Social Sciences, and Humanities (LaTeCH)*, pages 78–86, 2011.
- Lilia Rozanova and Alexander Temerev. A glyph is not a letter, a token is not a word, a space is not a space: What the units of Voynichese are not, 2026. arXiv:2608.17096 (v1, 17 August 2026).
- Gordon Rugg. An elegant hoax? A possible solution to the Voynich manuscript. *Cryptologia*, 28(1):31–46, 2004. doi: 10.1080/0161-110491892755.
- Dzianis Sachak. voynich-fingerprint, 2026. GitHub repository, <https://github.com/SachekDenis/voynich-fingerprint> (18 September 2026; MIT licence).
- Scott Schechter. voynich-decoded, 2026. GitHub repository, <https://github.com/scott-schechter/voynich-decoded>; version tested: last commit of 26 March 2026; no licence stated.
- Andreas Schinner. The Voynich manuscript: evidence of the hoax hypothesis. *Cryptologia*, 31(2):95–107, 2007. doi: 10.1080/01611190601133539.
- Emma May Smith. Linestart words, 2015. Agnostic Voynich (blog), 29 March 2015. <https://agnosticvoynich.wordpress.com/2015/03/29/linefirst-words/>.
- Emma May Smith. Proof of linestart transformation, 2016a. Agnostic Voynich (blog), 6 November 2016. <https://agnosticvoynich.wordpress.com/2016/11/06/proof-of-linefirst-transformation/>.
- Emma May Smith. What is [m]?, 2016b. Agnostic Voynich (blog), 27 December 2016.
- Emma May Smith. Last-first combinations and transformation theory, 2017a. Agnostic Voynich (blog), 14 August 2017.
- Emma May Smith. Initial [o] transformation, 2017b. Agnostic Voynich (blog), 16 May 2017.
- Emma May Smith and Marco Ponzi. Glyph combinations across word breaks in the Voynich manuscript. *Cryptologia*, 43(6):466–485, 2019. doi: 10.1080/01611194.2019.1596998.
- Tavi Stafford. Seven habits of highly eccentric paragraphs. In Colin Layfield and John Abela, editors, *Proceedings of the 1st International Conference on the Voynich Manuscript 2022 (VOY2022)*, volume 3313 of *CEUR Workshop Proceedings*, 2023. Paper 13.
- Jorge Stolfi. A grammar for Voynichese words, 2000. <https://ic.unicamp.br/~stolfi/EXPORT/00-EXPORT/00-06-07-word-grammar/>.
- tavie. Summary of Voynich day presentation on line patterns, 2024. Voynich Ninja forum, thread 4343, 8 August 2024. <https://www.voynich.ninja/thread-4343.html>.
- Torsten Timm. How the Voynich manuscript was created, 2014. arXiv:1407.6639.

- Torsten Timm and Andreas Schinner. A possible generating algorithm of the Voynich manuscript. *Cryptologia*, 44(1):1–19, 2020. doi: 10.1080/01611194.2019.1596999.
- Johannes Trithemius. *Polygraphiae libri sex*. 1518.
- University of Arizona. Radiocarbon dating of the Voynich manuscript vellum (four samples, 1404–1438 at 95%), 2011. Press release, 10 February 2011; work of the NSF-Arizona AMS Laboratory led by G. Hodgins. Account in R. Zandbergen, <https://www.voynich.nu/extra/carbon.html>.
- Siv Bugge Vatne. Cracking the Voynich cipher, 2021. Self-published PDF, October 2021. https://sivbuggevatne.com/wp-content/uploads/2021/10/verson_1_75mb_2.pdf.
- Elmar Vogt. The Voynich line: some statistical observations, 2012. Working paper, 27 November 2012. https://voynichthoughts.wordpress.com/wp-content/uploads/2012/11/the_voynich_line.pdf.
- Peter Wayner. Mimic functions. *Cryptologia*, 16(3):193–214, 1992. doi: 10.1080/0161-119291866883.
- Klaus-Peter Wegera, Hans-Joachim Solms, Ulrike Demske, and Stefanie Dipper. Reference corpus of early new high german (1350–1650), version 1.0.2, 2021.
- Whitehatnetizen. voynich-investigation (generators U2 and U3), 2026. GitHub repository, <https://github.com/Whitehatnetizen/voynich-investigation>, commit ccd5db0 (13 July 2026); MIT licence.
- René Zandbergen. The Cardan grille approach to the Voynich MS taken to the next level, 2021. arXiv:2104.12548.
- René Zandbergen. IVTFF – intermediate Voynich MS transliteration file format, 2025a. Document issue 2.0.2, 8 July 2025 (file format 2.0.1). https://www.voynich.nu/software/ivtt/IVTFF_format.pdf.
- René Zandbergen. Transliteration files of the Voynich manuscript: Zandbergen–Landini (ZL3b), Takahashi (IT2a), Glen Claston (GC2a), 2025b. ZL3b of 13 May 2025; IT2a and GC2a of 25 June 2025. <https://www.voynich.nu/transcr.html>. Public domain (CC0).
- René Zandbergen. Some special properties of labels in the Voynich MS, 2026. <https://voynich.nu/extra/labels.html> (updated 11 September 2026).
- Massimiliano Zattera. A new transliteration alphabet brings new evidence of word structure and multiple “languages” in the Voynich manuscript. In Colin Layfield and John Abela, editors, *Proceedings of the 1st International Conference on the Voynich Manuscript 2022 (VOY2022)*, volume 3313 of *CEUR Workshop Proceedings*, 2023. Paper 10.
- Zachary Ziegler, Yuntian Deng, and Alexander Rush. Neural linguistic steganography. In *Proceedings of EMNLP-IJCNLP 2019*, pages 1210–1215, 2019. doi: 10.18653/v1/D19-1115.
- Paula Zyats, Erin Mysak, Jens Stenger, Marie-France Lemay, Anikó Bezur, and David D. Driscoll. Physical findings. In Raymond Clemens, editor, *The Voynich Manuscript*, pages 23–37. Yale University Press and Beinecke Rare Book & Manuscript Library, New Haven, 2016.

A. Definitions of the main measures

Notation: a line is a sequence of words $w_1, \dots, w_n$; a word is a sequence of glyphs (Section 3); $\ell(w)$ and $f(w)$ are its last and first glyph. Plug-in entropies and mutual informations are in bits. Unless stated, a null is averaged over 50–1,000 permutations and z is the distance of the observed value from the null mean in null standard deviations.

Junction. $I = I(\ell(w_i); f(w_{i+1}))$ over all pairs of adjacent words in the same line; $E = I - \bar{I}_{\text{null}}$, the null shuffling the words of each line (same marginals and sample size, hence the same finite-sample bias). The variant with position (e3c87) shuffles $f(w_{i+1})$ among pairs with the same class of position (index of w_i in the line: 0, 1, 2, 3, 4 or more; whether w_{i+1} ends the line).

Mirroring of within-word rules. $\text{PMI}_{\text{in}}(a, b) = \log \frac{n_{\text{in}}(a, b) N_{\text{in}}}{n_{\text{in}}(a) n_{\text{in}}(b)}$ over consecutive glyphs inside words, and $\text{PMI}_{\text{x}}(a, b)$ likewise over pairs $(\ell(w_i), f(w_{i+1}))$; ρ is Spearman’s correlation over the cells with at least five counts in both tables.

Predictability of spaces. A rule $p(\text{space} \mid a, b)$ is estimated on the first half of the lines over all pairs of consecutive glyphs (a, b) (backing off to a , then to the global rate); on the second half a space is predicted when $p > 0.5$; the score is the F1 of the predicted spaces.

Attested wrong cuts. The glyph stream of each line is re-spaced at random with the rule above (five re-spacings); among the pieces of at least three glyphs that do not coincide with an original word, the share that are word types of the text (median over re-spacings).

Form filling and frequency–form correlation. A first-order glyph chain with start and end symbols (add-0.5 smoothing) is estimated on the word types. For each length $L = 3, \dots, 7$ the K_L most probable sequences are found by exact search, K_L being the number of attested types of that length; filling is the share of them that are attested. The frequency–form correlation is Spearman’s correlation, within length classes and pooled after centring the ranks, between the per-glyph log-probability of a type and the logarithm of its frequency (types occurring at least twice).

Agreement of choices (short state). For a choice c with forms coded $x \in \{0, 1\}$ (for example $k = 1, t = 0$), every inner word of a line of at least six words that shows the choice is an occurrence, with its *covered word* (the word with the variant position masked), hand, page and number of glyphs g written before it in the line. Its expected value is

$$p_a = \bar{x}_{\text{hand, covered}}^{(-a)} + (\bar{x}_{\text{page}}^{(-a, -b)} - \bar{x}_{\text{hand}}^{(-a)}) + \beta_c (g_a - \bar{g}_{\text{page}}^{(-a, -b)}),$$

clipped to $[0.01, 0.99]$, where the superscripts mean leave-one-out (or leave-two-out) means and β_c is the within-stratum slope of x on g . For pairs (a, b) of occurrences of the same choice at distance d words in the same line, excluding pairs whose covered words are equal or one edit apart,

$$K(d) = \frac{\sum_{(a, b)} [\mathbb{1}(x_a = x_b) - e_{ab}]}{\sum_{(a, b)} [1 - e_{ab}]}, \quad e_{ab} = p_a p_b + (1 - p_a)(1 - p_b).$$

Because p is estimated from the same data, K is biased in a way that depends on the text (by up to +0.37). The corrected value is $K_c(d) = K(d) - \bar{K}(d)$, where $\bar{K}$ is the mean of K over 50 shuffles of the values x among occurrences with the same hand, choice and covered word: the shuffles keep the bias and remove any real agreement. On synthetic texts the correction brings a text without agreement back to zero and leaves a true agreement unchanged (e3c48).

Drift along the line. Within strata (choice $\times$ covered word), the least-squares slope of x on g , pooled as $\sum \text{cov} / \sum \text{var}$ over strata, leaving out the first and last word of each line; reported per ten glyphs.

Copying from the lines above. For paragraphs of at least four lines and lines from the third on, the share of words that have an identical word, or one at one edit, in the two lines above; the null shuffles the order of the lines within the paragraph, so that the “lines above” are other lines of the same paragraph; E = observed – null mean.

Left margin. For the lines of a paragraph from the second on, with at least three words and a first word of at least two glyphs, the number of consecutive lines whose first two glyphs coincide, divided by its mean over 1,000 shuffles of the order of the lines within the paragraph.

Repetition. The share of adjacent pairs in a line that are the same word (words of at least two glyphs) or differ by one edit (words of at least three glyphs), and its ratio to the mean share after shuffling the words within each line (20 shuffles), or to $\sum_w p_w^2$.

Intervals. Contributions are summed per page; pages are grouped by bifolium (or quire), and groups are resampled with replacement 2,000 times; intervals are the 2.5th and 97.5th percentiles. For the junction the null is recomputed inside each resample from 20 within-line shuffles stored per page, so that observed and null share the same finite-sample bias.

Table 7: The main tests corrected for multiplicity: Holm within seven families, and Bonferroni over the 715 experiments logged before this test ($p < 7.0 \times 10^{-5}$, $|z| > 3.97$). Experiment e3c92.

family	test	statistic	Holm	Bonferroni
junction	junction between adjacent words	$z = 191.7$	yes	yes
	<i>qo</i> -/ <i>o</i> - after the previous word	$z = 19.4$	yes	yes
	- <i>l</i> /- <i>r</i> before the next word	$z = 12.7$	yes	yes
	look-ahead, end of the first word changes	$p = 0.0002$	yes	no
	look-ahead, next word adapts	$p < 0.0001$	yes	no
line edges	- <i>m</i> at line end; - <i>r</i> drops	$z = 26.7; -13.6$	yes	yes
	<i>y</i> -, <i>s</i> - at line start	$z = 16.6; 16.0$	yes	yes
copying	beyond the paragraph’s vocabulary	$z = 17.4$	yes	yes
left margin	<i>qo</i> -, <i>o</i> - avoided below the same start	$z = -9.7; -7.0$	yes	yes
	<i>d</i> -, <i>ch</i> -, <i>y</i> - avoided	$z = -3.8; -3.6; -3.5$	yes	no
short state	adjacent; two–three words (by bifolium)	$z = 12.3; 8.1$	yes	yes
drift	<i>sh</i> / <i>ch</i>	$z = -6.3$	yes	yes
	<i>qo</i> / <i>o</i> , <i>k</i> / <i>t</i> , - <i>ey</i> / <i>-dy</i>	$z = -3.0; -3.9; -2.8$	yes	no
book	bifolium identity; languages A and B	$z = 6.6; 13.4$	yes	yes

B. The log of experiments

The repository contains one row per experiment from e01 to e3c92 (`white_paper/revision e/registro_esperimenti.csv`), with date, question, files, outcome against the preregistered criterion, deviation and notes, and a summary with the borderline classifications. The log was compiled semi-automatically from the preregistrations, results and notebook; every quoted outcome was checked to appear verbatim in the sources, and 110 borderline classifications are listed for the reader to judge.

series	confirmed	refuted	mixed	inconclusive	superseded	descriptive	not run	total
e01–e99	20	25	8	9	2	32	2	98
e100–e199	10	61	9	18	9	3	2	112
e200–e299	18	66	4	5	9	6	7	115
e300–e399	37	26	15	13	2	7	1	101
e3a	43	24	12	2	12	4	2	99
e3b	22	22	12	12	29	1	1	99
e3c	20	35	11	13	6	4	3	92
all	170	259	71	72	69	57	18	716

“Confirmed” and “refuted” refer to the affirmative answer to each preregistered question. Experiments e01–e28 were imported from earlier work and re-run, without preregistration. “Superseded” marks outcomes later withdrawn or reversed (in four cases only the method was corrected and the result upheld). Of the 80 declared deviations, 19 are outcomes read differently from the literal criterion; the most consequential is e384, whose criterion (set from the range of short, noisy texts) literally gave “the junction carries over the line break as in prose”, while the measurements show a closed line ($Q \approx 0$, against 0.49 in languages; e3c89).

C. Internal checks: provisional readings corrected by later work

During the research (before version 1).

1. The agreement measure was biased by up to +0.37 (Appendix A). Before the correction it had given the Anglo-Saxon scribe of the *Hatton Gospels* a “window” like the Voynich’s, made Volapük look similar to the Voynich, placed the shape of the Voynich’s agreement outside the range of languages (it is at the top), and suggested that the state travels only on occurrences of the choice.
2. The state of the choices first looked like a window of three words dropping to zero; it fades gradually (half-life about three words), and a slow component crosses lines.
3. A per-line “setting” of the choices, like a key per line, did not hold.
4. After the form-only variants of the Glen Claston transliteration we read the state as tied to words rather than to letter shapes; a real scribe with a window in a form-only choice made that generalisation too broad.
5. An uncorrected measure suggested that hand 1 lacks the state; corrected, all hands have it.
6. An apparent reset of the choices at the line break came from comparing with the page average (line edges have forms of their own).
7. The first left-margin test used a defective null; the corrected one confirms the result.
8. A share of the link between whole words attributed to the paragraph was a mechanical effect of smaller groups.
9. Rare forms seemed to be copied from the line above more than others; with the second transliteration only a trend remains.
10. Two formally positive reading candidates (a syllabic one and a German one) were artefacts: the first came out the same on the shuffled Voynich, the second changed sign with the starting points of the search.

11. Three criteria called “absent” an effect whose interval only touched zero; the estimates were positive and similar to the rest. Since then “absent” requires the upper end of the interval below half the reference value.

Prompted by the external review of version 1.

12. “71 languages” were 71 texts, 15 of them in constructed languages; per natural language almost all comparisons hold, but two maxima came from Toki Pona, the junction is exceeded by two languages of 24 (not “six of 71”), and the per-text junction values of version 1 came from an unsaved exploratory check with a different glyph segmentation (e3c84).
13. The “weak space inside a chain of glyphs” does not survive a word-grammar null: predictable spaces, attested wrong cuts and form filling follow from word form (e3c85). The junction and its mirroring of within-word rules survive.
14. “Repeats the previous word more than chance” becomes “does not avoid repeating it, and repeated words cluster in the line” (e3c88); “no comparison text has this signature” for the decline of repetition within the line was wrong (two languages of 20; e3c84).
15. The drift was said to be “three to fifteen times” that of the Nordic scribes; relative to the strongest scribal drift it is about 2.5 to 4.5 times.
16. “Not found in any of 79 scribes” was too strong for the short state: with adequate power it is absent in 14 of 29 choices, and Holm A 10 shows a weaker state-like agreement (e3c90, e3c91). In the first run of e3c90 the drift intervals of the Nordic scribes were read in the wrong unit (per glyph instead of per ten glyphs), giving 28 instead of 20 choices “absent with power”; corrected the same day.
17. Version 1 described an early version of the voynichizer (message in the spelling choices); the published version carries it in the word counts.
18. Version 1 presented as new several regularities described before (Section 2).

D. Glossary

EVA conventional alphabet to transliterate the Voynich into Latin letters; it does not indicate sounds.

glyph a sign of the manuscript; *ch*, *sh* and the bench gallows count as one.

gallows the tall signs *k*, *t*, *p*, *f*.

Currier languages A and B the two varieties of writing into which the pages divide.

junction information that the last glyph of a word gives about the first glyph of the next, beyond chance.

junction rule a rule linking the form of a word to its neighbour (sandhi-like).

variant forms pairs of words that differ by one glyph in a fixed position (for example *k/t*).

short state the tendency to repeat the same variant for a few words.

scorecard the 18 properties with pass bands that the generator must reproduce.

judge a classifier retrained on each generated book to tell its pages from the real ones; its AUC is a classifier two-sample test.

AUC area under the ROC curve: 0.5 for guessing, 1 for perfect separation.

arithmetic decoder the inverse of an arithmetic coder; fed with random bits it samples from a probability model.

preregistration a dated description, written before running, of question, data, measure and criterion.

95% interval a bootstrap percentile interval: the range of the middle 95% of the estimates over resamples of bifolia (or pages).

E. Data sources and licences

data	source	licence
Voynich transliterations ZL3b, IT2a, GC2a	Zandbergen (2025b)	public domain (CC0); in the repository
71 meaningful texts, 38 gibberish texts	Gaskell and Bower (2023), released subset	modified MIT (cite the paper)
Bible in 100 languages	Christodouloupoulos and Steedman (2015)	CC0
Latin Library; Divinum Officium	CLTK collection; Divinum Officium project	as stated by the sources (MIT for Divinum Officium)
six Nordic manuscripts	Menota; each text with the editor and version in its header	CC BY-SA 4.0
73 German manuscripts	Wegera et al. (2021), v1.0.2	CC BY 4.0 (Zenodo) or CC BY-SA 4.0 (file); we follow the latter
Copiale	Knight et al. (2011a,b), project page at Stockholm University	research use
Naibbe	Greshko (2025), repository greshko/naibbe-cipher	modified MIT
Timm and Schinner generator	Timm and Schinner (2020)	MIT
U2, U3	Whitehatnetizen (2026)	MIT
readings tested	Bax (2014); Cheshire (2019); Gatta (2026); Schechter (2026); Vatne (2021)	as published; Schechter: none stated, local copy not redistributed